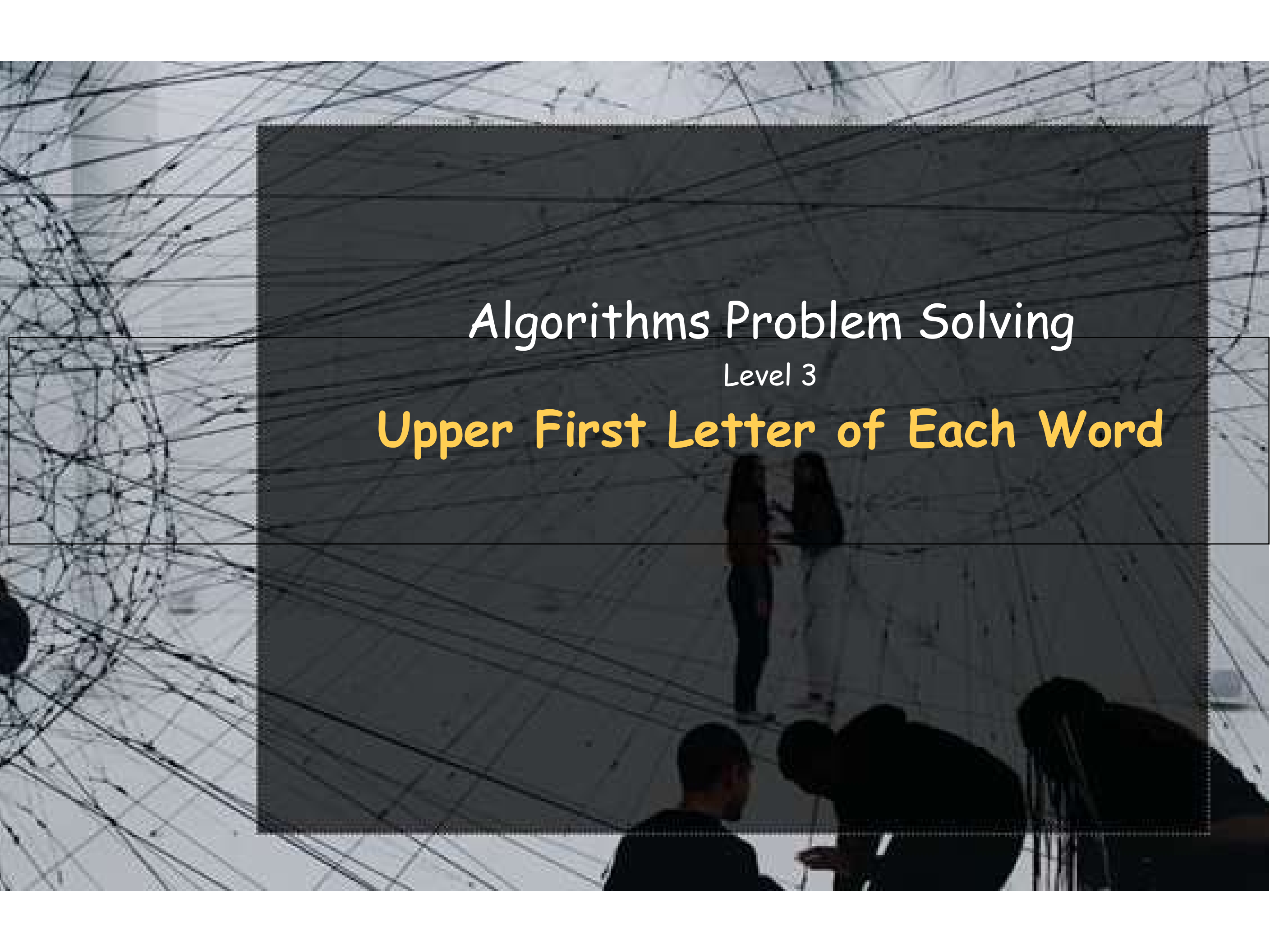


# Algorithms Problem Solving Level 3

Upper First Letter of Each  
Word



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## Upper First Letter of Each Word

# Definition

## Word-Initial Capitalization

-  Program converts the first letter of every word to uppercase.
-  Characters after the first letter of each word remain unchanged.

# Explanation

## Word Start Detection

-  A boolean state such as `isFirstLetter` identifies when the next alphabetic character begins a word.

-  The state becomes active at the beginning of the string and after whitespace.

## Character Conversion

-  When the first character of a word is detected, `toupper()` converts it to uppercase.

-  Afterward, the state changes so subsequent characters are not modified.

## String Mutation

-  The original string can be modified directly by changing individual characters.

-  Alternatively, the function can return a modified copy.

# Procedures

1. Read the input string.
2. Initialize the word-start state.
3. Traverse the string.
4. Convert each word's first letter using ``toupper()``.
5. Update the state after processing each character.
6. Output the modified string.

# 💡 Example

Input:

sara alterkawi

➡ Result

Sara Alterkawi

## Note

- ◆ Existing uppercase and lowercase letters after the first character remain unchanged.
- ◆ ``toupper()`` operates on individual characters.
- ◆ This operation is similar to a limited form of title formatting, not full language-aware title casing.

## ⊘ Warning

- ◆ ``toupper()`` does not directly accept a complete ``std::string``.
- ◆ Non-alphabetic characters at the start of a token are not converted into letters.
- ◆ For portable use with ``<cctype>``, character values should be handled appropriately when passing signed ``char`` values.



# Comparison

## ▨ `UpperFirstLetterOfEachWord()`

- ✓ Changes only the first letter of each word.
- ✓ Preserves the remaining characters.
- ✓ Produces word-initial capitalization.

## ▤ `UpperAllString()`

- ✓ Converts every applicable letter to uppercase.
- ✓ Changes internal letters as well.
- ✓ Produces an all-uppercase string.

# ✦✦ Summary

- ✓ Detects the first character of each word.
- ✓ Converts applicable word initials to uppercase.
- ✓ Preserves the rest of the original string.
- ✓ Demonstrates character-level string transformation.

Thank you 😊